Vertical Integration

Session 20

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Business vs. Corporate Strategy

Strategic Management

Business Strategy

How does a firm compete in a given industry?

- **Competitive strategy** and **Positioning**
- **Resource-based view**

Corporate Strategy

is concerned with *WHERE* a firm competes.

Which Industry or Segment, but also about the *scope of activities*.

Corporate Strategy

Where does a firm compete → boundaries of the firm

Scope of the firm:

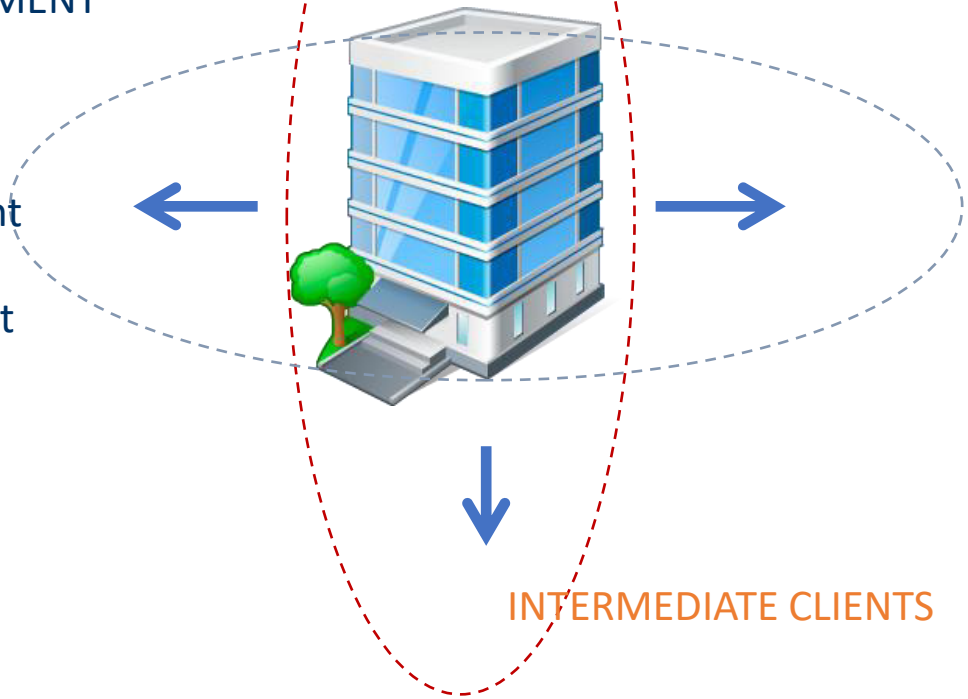
1) **Vertical** boundaries (Vertical integration)
2) **Horizontal** boundaries (Diversification)
3) **Geographical** boundaries (Internationalization)

VERTICAL INTEGRATION

SUPPLIERS

HORIZONTAL DEVELOPMENT

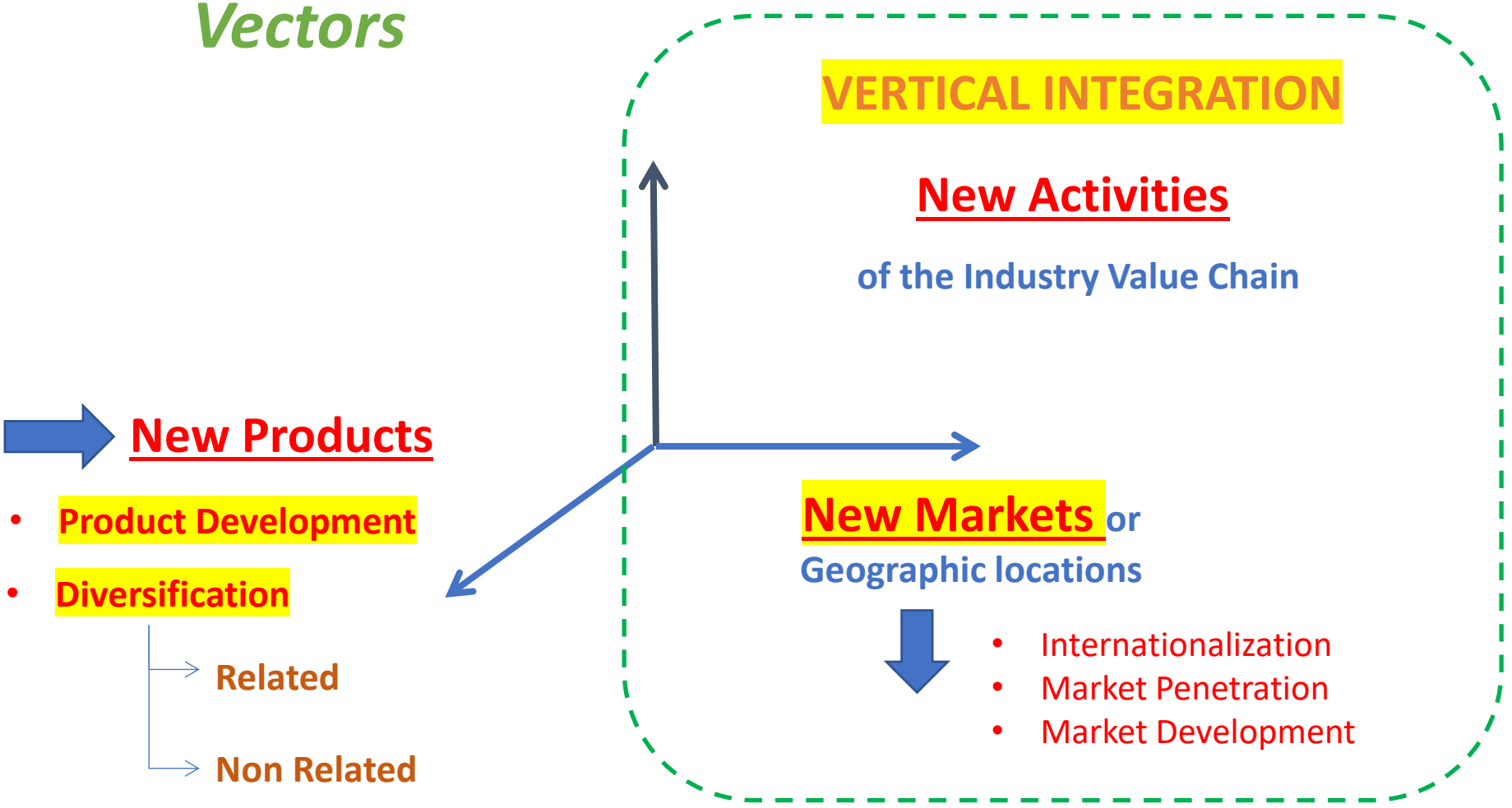
- Diversification
- Product development
- Market penetration
- Market Development
- Internationalization
- Etc.



INTERMEDIATE CLIENTS

✓ Organic
✓ Inorganic

Growth Vectors



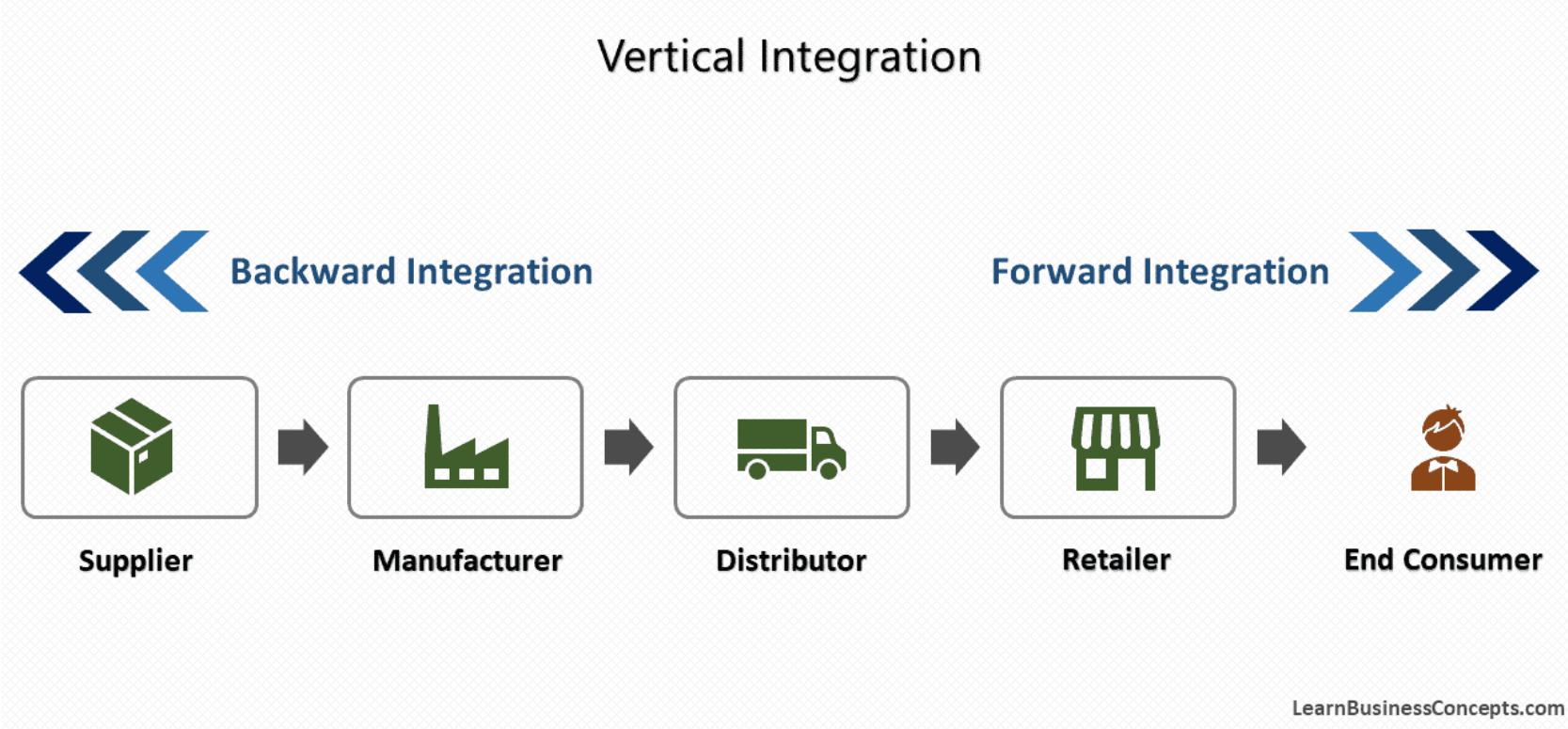
1. Vertical Integration

2. Transaction Costs & Administrative Costs

3. Motives for Vertical Integration

4. Types of Vertical Relationships

5. Strategic Implications


Vertical Integration
Backward Integration
Forward Integration
Supplier
Manufacturer
Distributor
Retailer
End Consumer
LearnBusinessConcepts.com

Forward and backward integration

Backward integration or Upstream

Forward integration or Downstream



Dairy farm



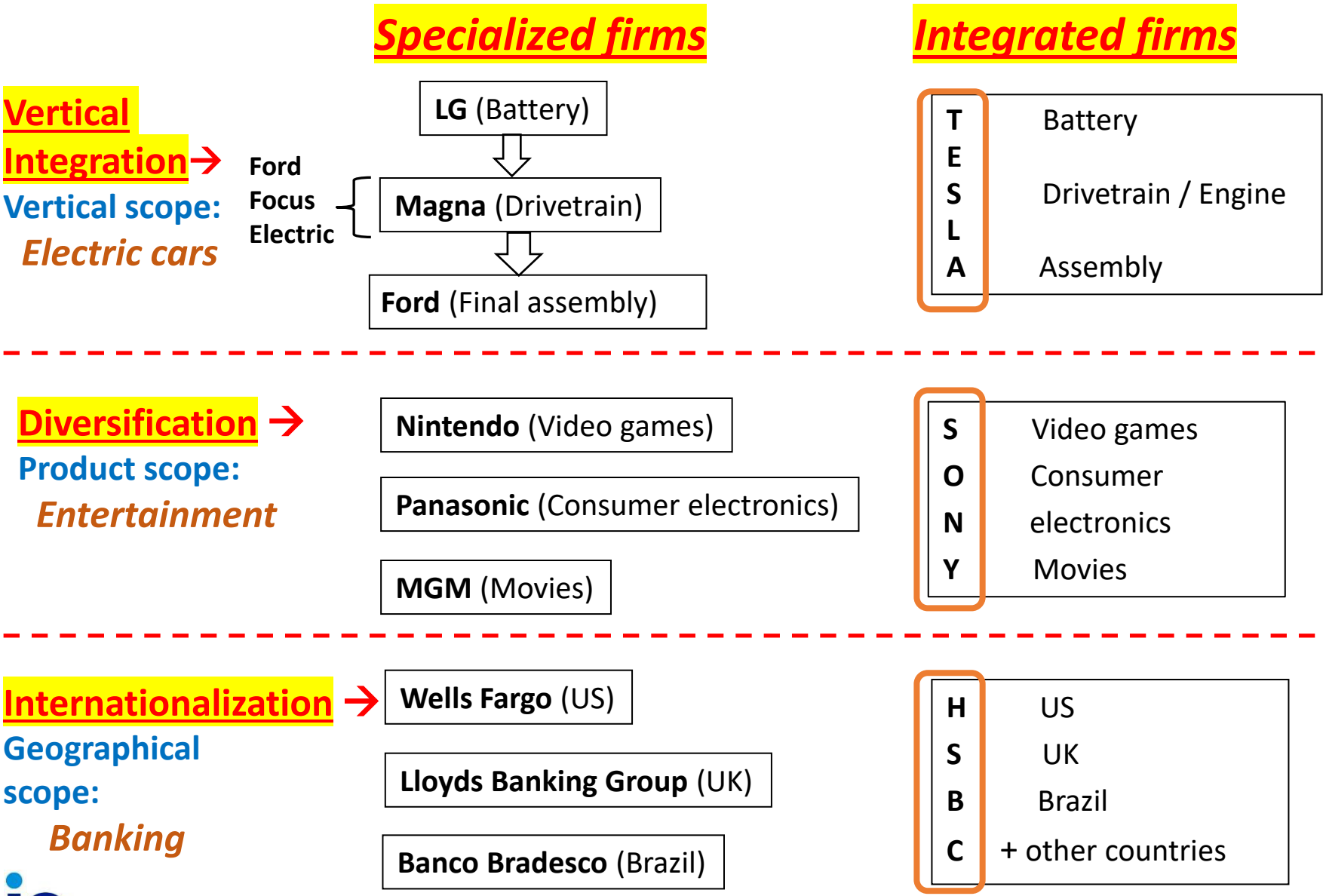
Ice cream maker



Ice cream shop

- **Backward Integration (Upstream integration)** = **Doing the activities of your supplier(s)** yourself / acquiring an upstream supplier
- **Forward Integration (Downstream integration)** = **Expanding business activities to customers**' activities / acquiring a downstream firm

Coffee Vertical integration explained (3m) https://www.youtube.com/watch?app=desktop&v=RbMp9blfvuU

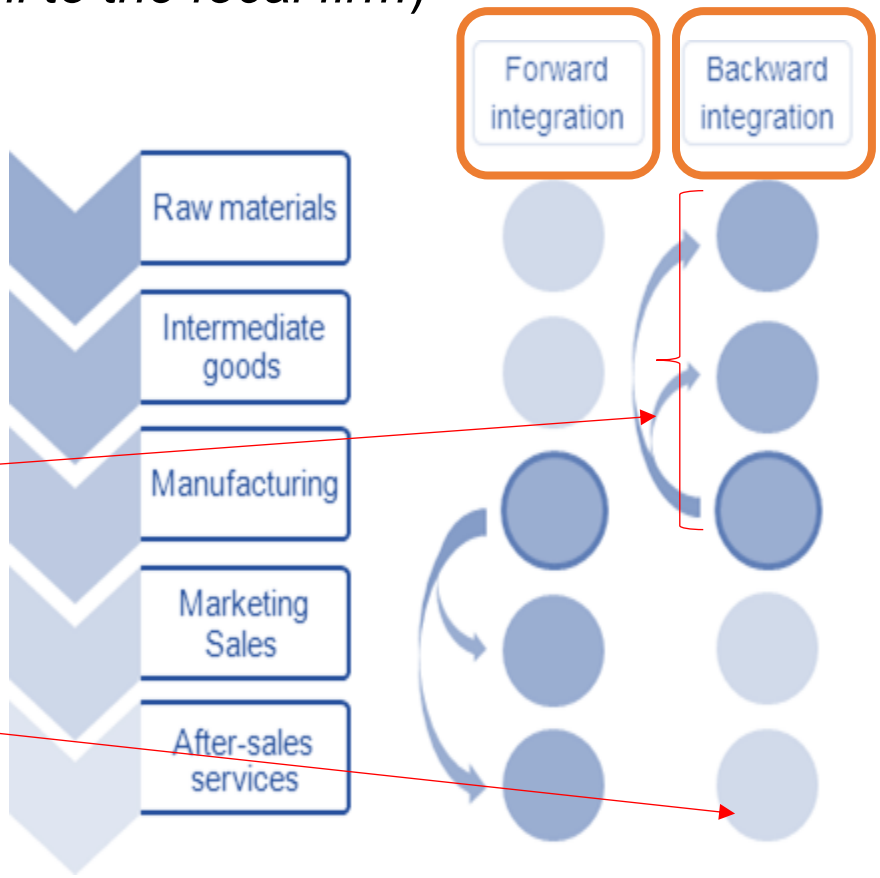

Specialized firms
Integrated firms
Vertical Integration
Vertical scope:
Electric cars
Ford Focus Electric
LG (Battery)
Magna (Drivetrain)
Ford (Final assembly)
TESLA
Battery
Drivetrain / Engine
Assembly
Diversification
Product scope:
Entertainment
Nintendo (Video games)
Panasonic (Consumer electronics)
MGM (Movies)
SONY
Video games
Consumer electronics
Movies
Internationalization
Geographical scope:
Banking
Wells Fargo (US)
Lloyds Banking Group (UK)
Banco Bradesco (Brazil)
HSBC
US
UK
Brazil
+ other countries

- The **vertical boundaries of a firm** define the activities that the firm performs itself (*as opposed to activities performed by other, independent firms that buy from/sell to the focal firm*)

- **Vertical Integration** = Company controls two or more stages of the supply chain

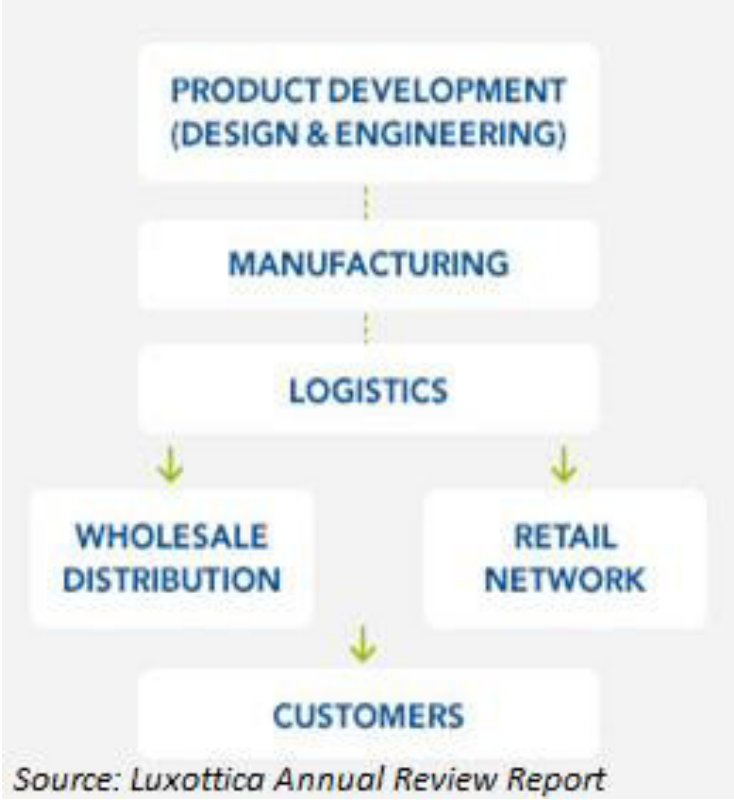
Several knots = vertically integrated firm

one knot = specialized firm



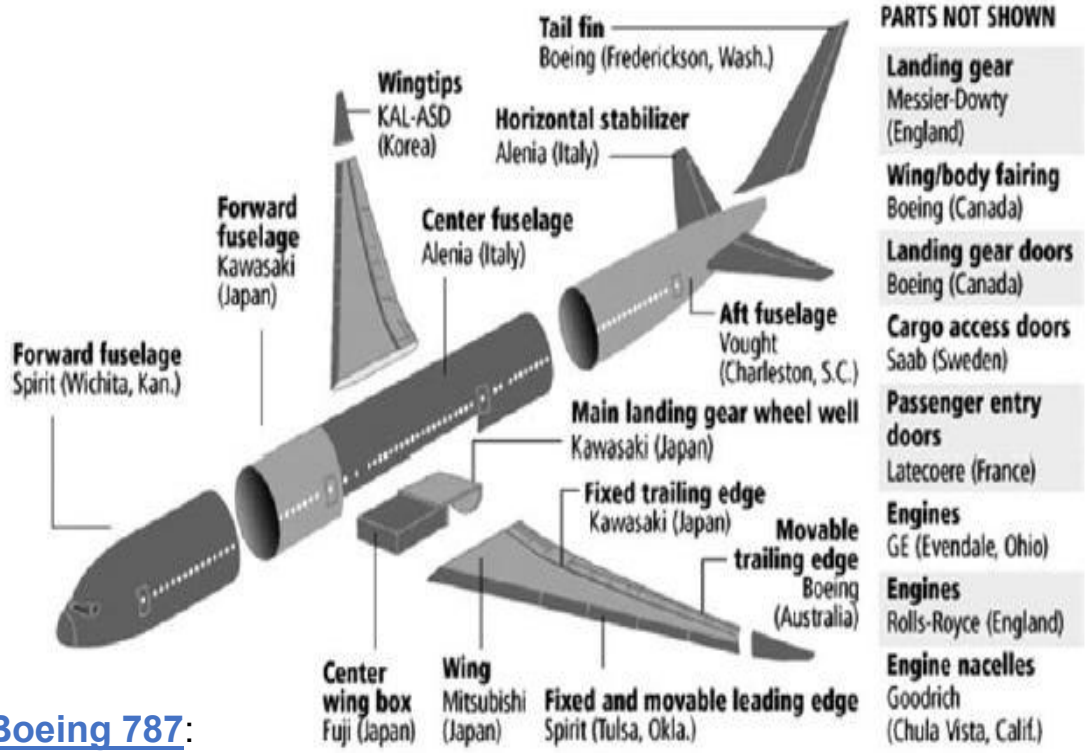
Luxottica Example

Vertical Integration



Source: Luxottica Annual Review Report

Example: Power and control – Boing's supply chain



No vertical Integration

Boeing 787's problems:

- Flaws of electrical systems
- Poor quality components and scarce oversight of suppliers

Boeing 787:

- Unconventional supply-chain management to reduce development costs and time
- **Boeing tried to act as a system integrator of parts produced by thousands of suppliers**



1. Vertical Integration

2. Transaction Costs & Administrative Costs

3. Motives for Vertical Integration

4. Types of Vertical Relationships

5. Strategic Implications

Transaction costs economics

- Pioneered by **Ronald Coase (1937)** and further developed by **Oliver Williamson (e.g. 1975; 1985)**
- Both received a Nobel prize





Two forms of economic organization

Market mechanism

- **Individuals and firms make independent decisions to buy and sell goods/services, based on market prices**
- *Characterized as the invisible-hand (Adam Smith), as coordination does not require conscious planning*
- ***Production is organized through market transactions***

Administrative mechanism

- **Decisions concerning production and resource allocation are made by managers and carried out within firms' hierarchies**
- *Characterized as the visible-hand (Alfred Chandler), as it involves active planning & direction*
- ***Production is organized inside the firm***

Both mechanisms create costs

↓ **Transaction costs**

↓ **Administrative costs**

Transactions costs

- **Transaction costs** → Costs of **organizing production activities via market contracts** with specialized firms.
- **Sources Transaction costs:**
  1. costs of searching market counterparties
  2. costs of negotiation
  3. costs of writing a contract
  4. costs of monitoring the contract
  5. costs of litigating (lawyers) and enforcing the contract

Administrative costs

- **Administrative costs** = costs of **organizing production activities inside an integrated firm**.

- **Sources of Administrative costs**

1. Managing different businesses (and eventual conflicts)
2. Differences in optimal scale between different stages of production.
3. Possible differences in capabilities needed at each stage of production
4. Lack of flexibility
5. Competition with customers and/or suppliers.
6. Compounding of risks.

Transaction costs economics

- What determines **which activities** are undertaken **within the firm** & which through **market contracts**?

The **relative cost** of organizing activities within the firm as compared to organizing across markets (Ronald Coarse)

Administrative costs > **Transaction costs** → **Market (BUY)**

Transaction costs > **Administrative costs** → **Within the Firm (MAKE)**

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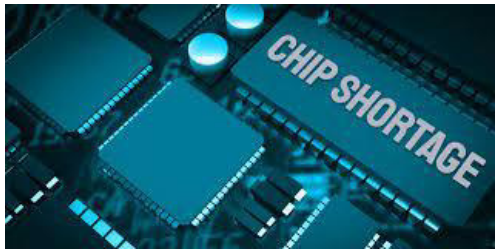
Vertical integration allows firms to avoid transaction costs

1. **Makes repeated transactions more efficient** Ex.: INDITEX (fast fashion)

2. **Reduces market uncertainty** Ex.: OIL Industry (Control of Resources)

Impossible to write a complete contract that covers every possible eventuality.

3. **Decreases Opportunistic behavior**



Important if transaction were asset specificity is high

→ Parties depend on one another.

→ Risk of opportunistic behavior (*i.e.,* ***one party can threaten the other with retaining supplies or raw materials.***) e.g.: *Why there is a global **semiconductor shortage**, how it started, who it is hurting, and how long it could last*

Asset specificity → Physical and human investments that are specialized and unique to a transaction.
The investments in asset would only be valuable from their use of the specific transaction (Williamson 1975, 1985, 1986). This means how specific the investment is for the activity and the costs required reallocating it for another use.

Motives for vertical integration

Benefit-related

Increase value for customers:

1. Control of the **product quality**
2. Allow for **asset specificity**

(e.g. Starbucks´s coffee quality)

3. Improved **time to market** (e.g.: Zara)

Value for customer → Starbucks:

- "vertical integration to the extreme" (Howard Schultz)
- The company buys and roasts all its own coffee and sells it through entirely company-owned stores

Efficiency-related

increase efficiency

4. Better integration of **production processes**.
5. **Reduce transaction costs**

- **Costs of coordinating market transactions** along the supply chain
- **Costs associated with sellers' or buyers' power**

The BENEFITS of Vertical Integration

Avoids transactions costs of market **contracts when:**

1. Few suppliers or small numbers of firms. E.g.: Design of microchips (ARM)
2. Transaction-specific investments E.g.: Software to coordinate with supplier
3. Opportunism and strategic misrepresentation E.g.: Microchips
4. Taxes and regulations on market transactions E.g.: Import taxes
5. Integrated Processes. Technical economies from integrating processes, but doesn't necessarily require common ownership

   E.g.: Car Industry or Iron and Steel production

6. Superior coordination. E.g.: Amazon delivery big cities

The COSTS of Vertical Integration

1. Difficult Managing different businesses

2. Differences between different stages of production

e.g. : The production of wood is lower that the production of furniture

3. Reduce the development of distinctive capabilities

e.g. : Not competing in the markets → Reduce power of getting new capabilities.

4. Incentive problems: lack of "high-powered" incentives. No competition.

5. Limits flexibility → in responding to demand fluctuations

→ in responding to changes in technology, customer preferences, etc.

(But, may be conducive to *system-wide flexibility)*

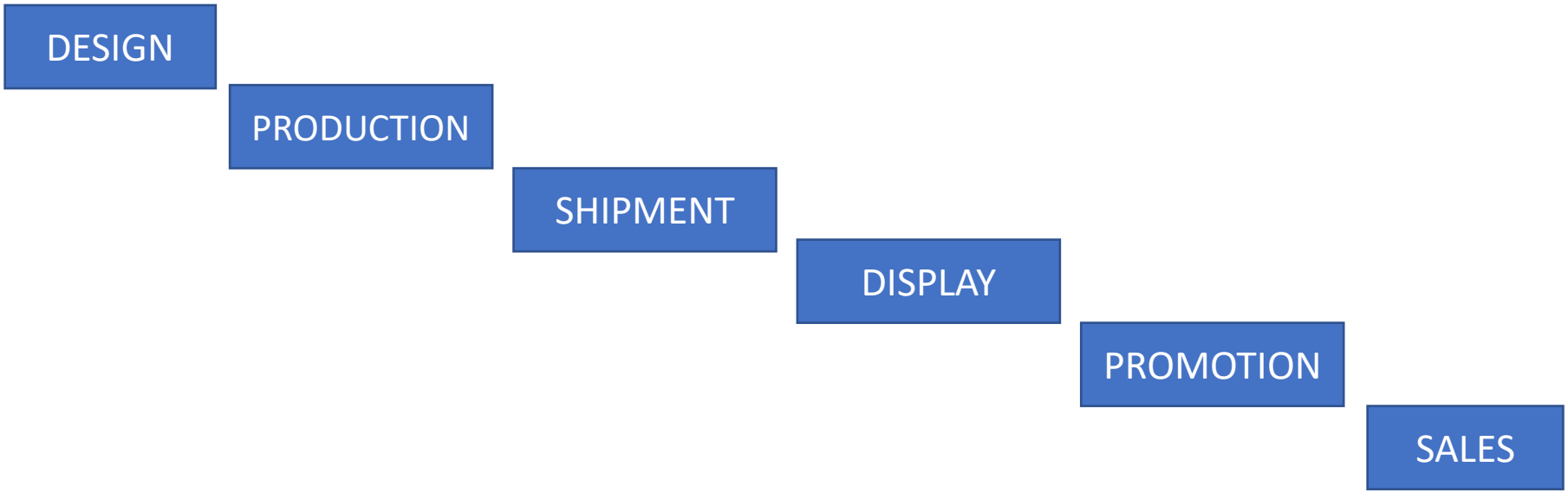
6. Increase Compounding risk. All risks concentrated in the same industry.

RISKS of Vertical Integration

1. Concentration of Resources in an Industry (Increase Global Risk)
2. Increased Exit Barriers.
3. Increased Organizational complexity.
4. The margin obtained by suppliers or intermediate customers, a not captured immediately.

Does ZARA "make" or "buy"?

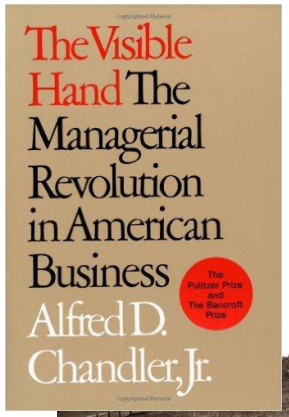
Zara is **vertically integrated**. It manages design, production, shipment, display, promotion, sales. *Everything, except the raw material cloth,* in which Zara has long term agreements with suppliers. … But what, if it were not?

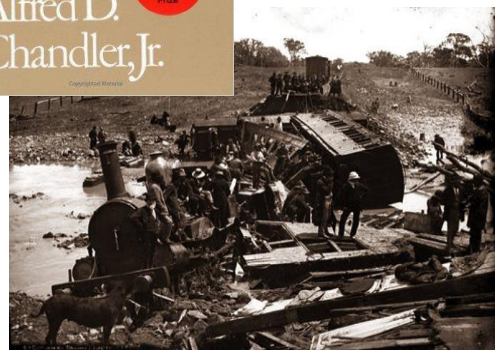


History and vertical integration

Before 70s - 80s → Growth phase (companies constantly grew in size)

- **Firms vertically integrated** in order to ensure constant flow of inputs (backward) and sufficient sales (forward)
- **Technological reasons for integration** (e.g. to increase railway safety, firms integrated conductors, station agents, locomotive engineers, etc.)





After 70s – 80s → Growth phase is reversed

- Trend towards **hybrid forms** **that combine vertical integration with market transactions**
- **Increase of outsourcing** and '**vertical partnerships' with suppliers** (e.g. Toyota's re-supplying system based on kanban cards)

TOYOTA

1. Vertical Integration

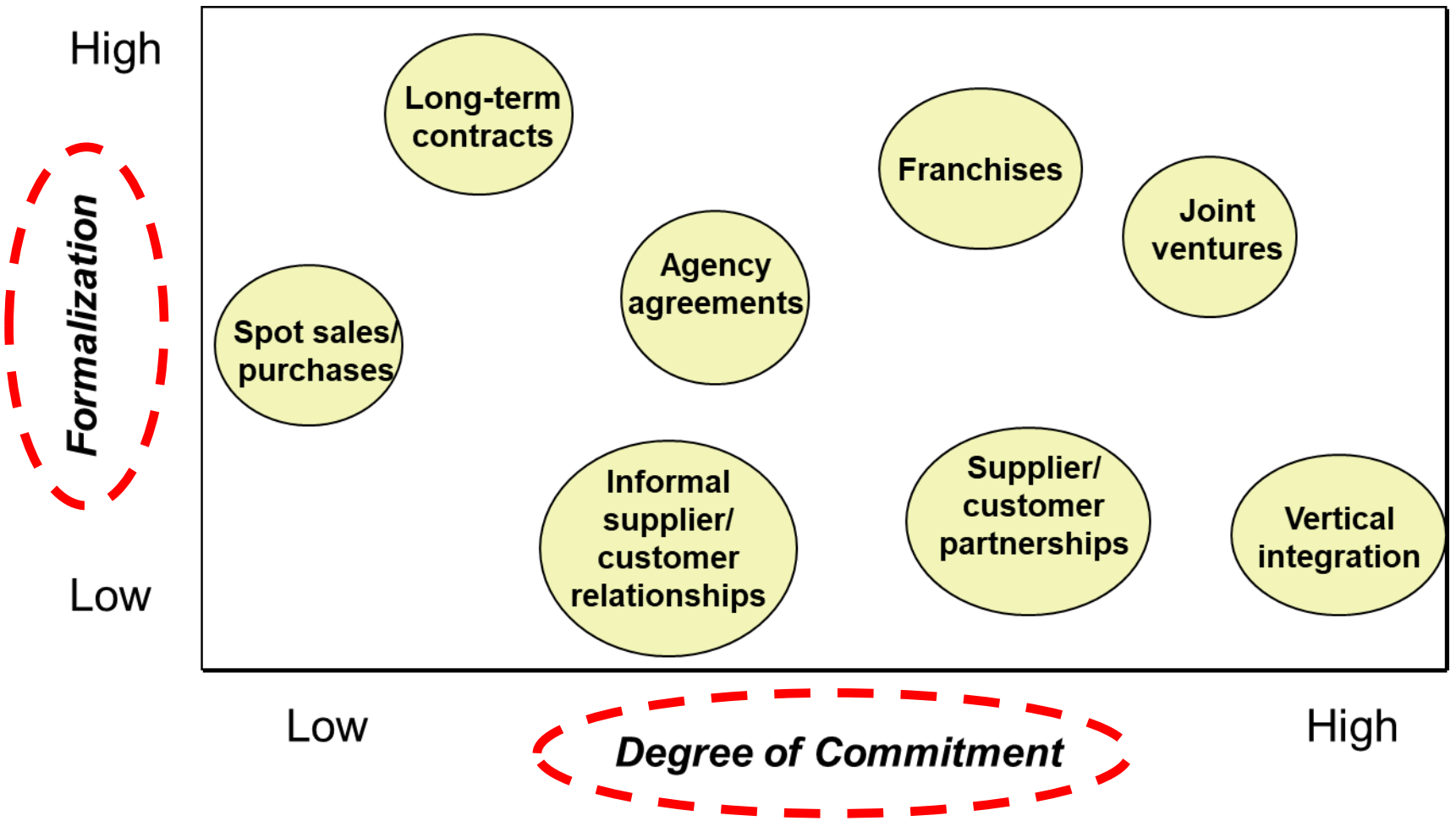
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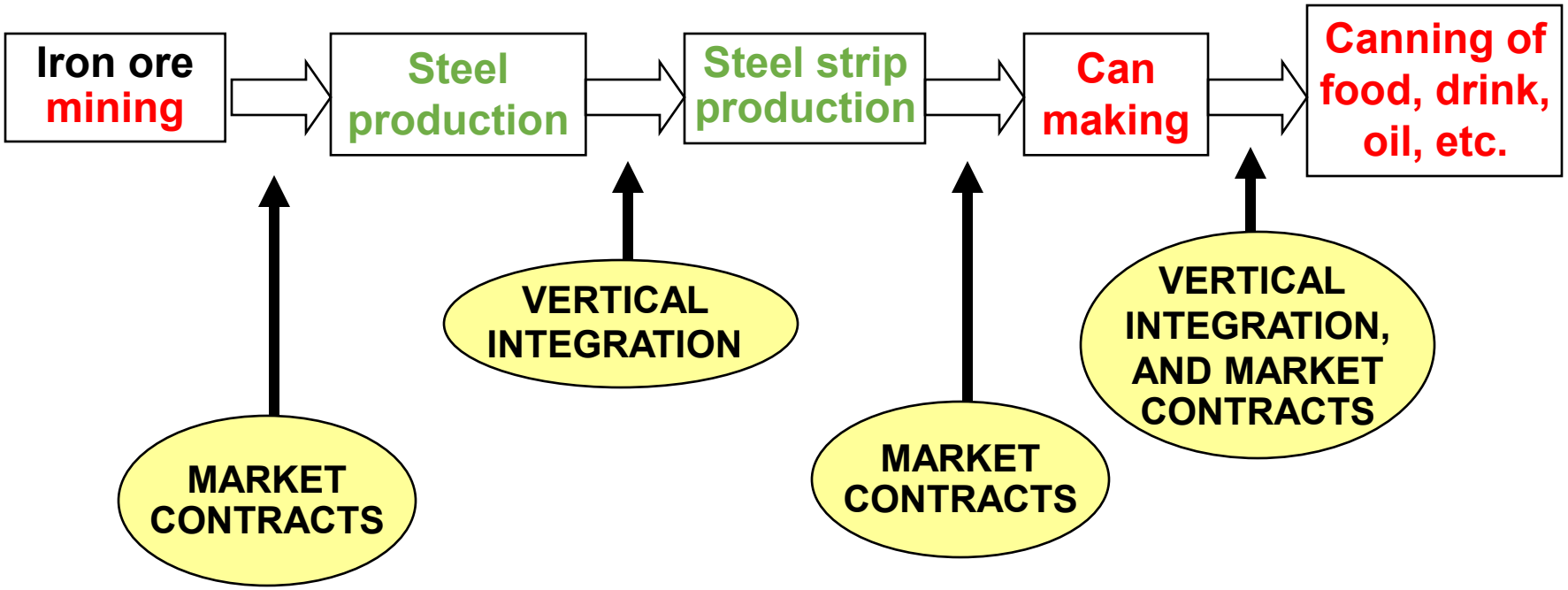
4. Types of Vertical Relationships

5. Strategic Implications

Types of vertical relationships



The Value Chain for Steel Cans



What factors explain why some stages are vertically integrated, while others are linked by market transactions?

A brief summary of trade-offs

	Market (Spot contract)	Types in-between	Vertical integration
Power/Control	None	Some	A lot
Flexibility	A lot	Some	None
Scalability	A lot	Some	None
Knowledge transfer / ease of communication	None	Some	A lot

Long-Term Contracts and Quasi-Vertical Integration

- Choices not limited to vertical integration or market contracts: →

Several intermediate types of vertical relationship: ***these may combine benefits of both market transactions and internalization***.

Key issues in designing vertical relationships:

- **No generic solution**: depends upon the resources, capabilities and strategy of the individual firm.
- **How is risk** to be allocated between the parties?
- Are the **incentives appropriate**?

Recent Trends in Vertical Relationships

1. From *competitive contracting* to *Supplier partnerships*, e.g. Car Industry
2. From vertical integration to Outsourcing (not just components, also IT, distribution, and administrative services).
3. Diffusion of Franchising
4. Technology partnerships (e.g. IBM- Apple; Canon- HP)
5. Inter-firm networks → A group of related organizations that partner and/or cooperate with each other in order to provide expanded products and services. (*e.g.: Toyota & Honda; BMW & Mercedes-Benz; One world alliance airlines*)

General conclusion: Boundaries between firms and markets are becoming increasingly blurred (less clear the boundaries).

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- Vertical integration is neither good nor bad

Optimal degree of vertical integration depends on Vertical integration checklist

Low Concentration of competitors

1. How many firms are in the vertically adjacent activity?

- The greater the number of firms the less advantageous is VI

Control of Resources

2. Asset Specificity. Do transaction-specific investments need to be made by either party?

- The greater the need for transaction-specific investments, the greater the advantages of VI

Control of Information

3. How is the **distribution of information** between the firms at each stage?

- The greater are **information asymmetries**, the more likely is opportunistic behavior and the greater the advantages of VI

Control of uncertainty

4. How **great is uncertainty** over the period of the relationship?

- The greater the uncertainty, the more incomplete is the contract and the greater the advantages of VI

If they are not similar scale production

If need capability development

If need high profit incentives

If unpredictable demand

If great risks at each stage

5. How similar are two stages in terms of the optimal scale of the operation? How strategically similar are the stages?

- The greater the dissimilarity, the less advantageous is VI

6. How critical is the continual upgrading of capabilities in the adjacent activity?

- The greater the need for capability development, the greater the disadvantages of VI

7. How important are profit incentives to performance in the adjacent activity?

- The greater the need for high-powered incentives, the greater the disadvantages of VI

8. How uncertain is market demand?

- The more unpredictable is demand, the less advantageous is VI

9. Does VI transmit risk between stages?

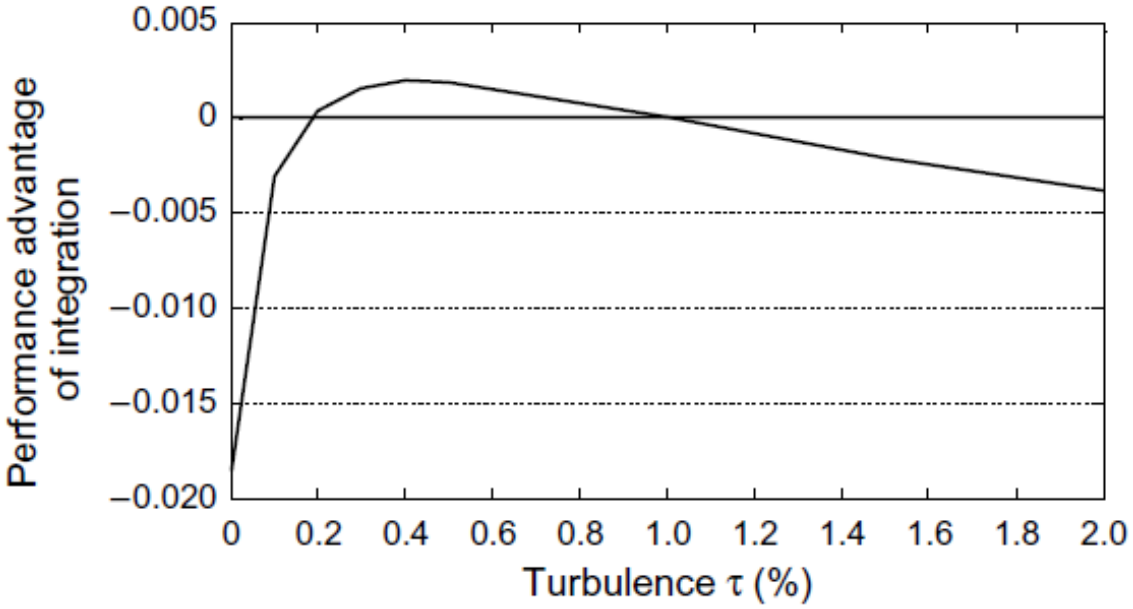
- The greater are risks at each stage, the more VI compounds risk

Vertical Integration v. Outsourcing: Key Considerations

Characteristics of the vertical relationship	*Implications for Vertical Integration (VI)*
How many firms in the adjacent stage?	The fewer the number, the less advantageous is VI
Do transaction-specific investments necessary?	The greater the need for transaction-specific investments, the greater the advantages of VI
Is information evenly distributed across the stages?	The greater are information asymmetries, the greater the advantages of VI
Is there uncertainty over the period of the relationship?	The greater the uncertainty, the more incomplete is the contract and the greater the advantages of VI
How similar is optimal scale between the two stages?	The greater the dissimilarity, the less advantageous is VI
How strategically similar are the two stages?	
Do capabilities in the adjacent stage need to be continually upgraded?	The greater the need for capability development the greater the disadvantages of VI
Are profit incentives critical to performance?	The greater the need for high-powered incentives the greater the disadvantages of VI
Is market demand uncertain?	Unpredictable demand reduces advantages of VI
Is the adjacent stage highly risky?	VI tends to compound risk

Turbulence and vertical integration

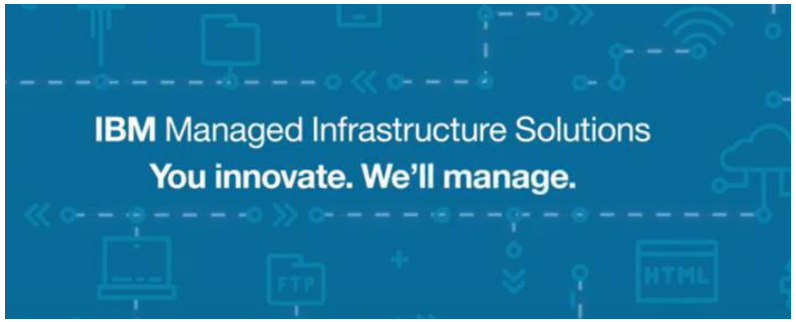
- Turbulence = technological change
- Integration becomes more attractive for moderate levels of turbulence
- Integration becomes less attractive for high levels of turbulence



Claussen, J., Kretschmer, T., & Stieglitz, N. (2014). Vertical scope, turbulence, and the benefits of commitment and flexibility. *Management Science*, *61*(4)

Solution providers

- Recent trend of forward vertical integration of technology firms
- Firms try to become **solution providers**
  - They provide both **services** and **products**
  - Take over some of the customer's activities



Advantages:

- Firms establish long-term relations
- Lock-in customers
- More continuous, stable stream of revenues

Assignment for next session:

Please read the case on **Lao Coffee** (Accessible via syllabus)

Questions to focus on while reading:

1. What is the type of Lao Coffee's vertical integration?
2. What motivates Lao Coffee's vertical integration?
3. What are the implications of Lao Coffee's vertical integration?
4. Is this a Social business model? How does the vertical integration helps social purposes?



Questions for Lao Coffee Case study

1. **What is the case about?**
   a) How is coffee made?
   b) What Resources and Capabilities did Say leverage during the formation process? Apply VRIO analysis to evaluate the strategic importance of them.
2. **Business Model Design, Vision and Mission.**
   a) What are the vision and mission of Bolaven Farms?
   b) What are the main characteristics of the Bolaven Farm business model?
3. **Vertical Integration**
   a) Why did Say choose vertical integration?
   b) What were the gains and pains of Lao Coffee's vertical integration?
4. **The Role of Social networks and social Business Models**
   a) How did the network of relations evolve?
   b) What impact did Say´s social network have created?
5. **Strategic Consultancy and Advises for the future**
   a) What would you have advised Say to do? (alternative solution)
   b) What are the lessons of the case? (vertical integration specifically)

Thanks a lot!

Any questions?

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